MIL-C-81175D(YD)
7 February 1983
SUPERSEDING
MIL-C-81175C(YD)
28 March 1977

MILITARY SPECIFICATION

CRANES, HYDRAULIC, WHEEL MOUNTED, 4 X 4, FULL REVOLVING, 15-TON CAPACITY, DED

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a diesel-engine-powered, self-propelled, four-wheel drive, four-wheel steer, full revolving, hydraulically operated crane of 15-ton capacity for torpedo, electronic gear, missile, or ammunition handling, when equipped with suitable attachments and used for construction or maintenance.

1.2 Classification. Cranes shall be one of the following models as specified (see 6.2).

Model I - Torpedo handling 15-ton crane.
Model II - General purpose 15-ton crane.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified (see 6.2), the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3810

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SPECIFICATIONS

MILITARY

MIL-V-173 - Varnish, Moisture- and Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-C-3580 - Crane and Crane-Shovels, Truck, Crawler, and Wheel-Mounted, Full-Revolving, and Their Attachments, Packaging of.
MIL-W-4088 - Webbing, Textile, Woven Nylon.

STANDARDS

MILITARY

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR (DoL)

OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies of DoL and OSHA should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards and Regulations.
Motor Carrier Safety Regulations.

(Application for copies of DoT should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents that are indicated as DoD-adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN NATIONAL STANDARDS INSTITUTE, INC., (ANSI)

B30.15 - Safety Standards for Mobile Hydraulic Cranes.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

MOBILE HYDRAULIC CRANE STANDARDS

PCSA Standard No. 2.

(Application for copies should be addressed to the Power Crane and Shovel Association, Marine Plaza, Suite 1700, 111 East Wisconsin Avenue, Milwaukee, WI 53202.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC., (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.

T&RA Yearbook.

(Application for copies should be addressed to The Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The crane shall consist essentially of a four-wheel drive, four-wheel steer, pneumatic-tired, diesel-engine-powered carrier, with rear end pintle hook and other mounted attachments. The carrier shall be equipped with a center-mounted, full-revolving telescopic boom of not less than 45-foot extended length and capable of not less than 40-foot radius. The maximum overall width of the crane shall be not more than 108 inches. The overall height, with crane boom in travel position, shall be not more than 142 inches. The vehicle weight shall be not less than 36,600 pounds (lb). Model 1 torpedo-handling crane, shall contain a nonrotating, forward-mounted operating cab and two torpedo cradles attached to the carrier as specified in 3.14. Model 2 general-purpose crane shall have a rotating operating cab, but no torpedo cradles. Model 2 shall be used as a general purpose crane or may be used in conjunction with truck or trailer transportation of torpedoes. The crane shall comply with PCSA Standard No. 2.

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3.2 First article. When specified (see 6.2), the contractor shall furnish a crane for first article inspection and approval (see 4.2.1 and 6.3).

3.3 Standard commercial product. The crane shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features that are not specifically prohibited by this specification, but are a part of the manufacturer's standard commercial product, shall be included in the crane being furnished. A standard commercial product is a product that has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects that would adversely affect the performance or maintainability of individual components or the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials that have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Maintainability. The crane design, together with component and accessory location and installation, shall permit ready accessibility to all items requiring periodic maintenance service in the field, which will be accomplished with the use of conventional general-purpose tools associated with equipment of this nature. The replacement and adjustment of components and accessories shall be accomplished with minimum drainage requirements and minimum disturbance to other elements of the crane.

3.7 Safety. The crane shall comply with OSHA regulations and ANSI B30.15. All rotating or reciprocating parts and all parts subject to high operational temperatures, that are of such nature or are so located as to be or become a hazard to the operating or attending personnel, shall be substantially guarded, or insulated, to the extent necessary to eliminate the hazard. The principal platform walking surfaces shall be an antiskid type. Ladders, steps, and handholds shall be provided in such quantity and of such size on the sides of the crane cab or superstructure that entrance thereto and exit therefrom may be unhampered and nonhazardous. Engine cooling fans shall have heavy grille or ring-type guards.

3.8 Stress requirements. Design of the completely assembled crane shall be such that the maximum unit stress imposed will not exceed the minimum strength margins contained in SAE J1063. Tests, in accordance with SAE J1063,

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shall have been performed on the first article being provided and test results shall be made available for Government inspection at the time of the first article tests (see 4.4.2).

3.9 Performance. The crane shall be capable of independently and simultaneously lifting and traveling with a sweep of 360° continuous rotation, and shall have a maximum rotational speed at a rate of not less than 2 revolutions per minute (rpm). The crane, without load, shall be capable of attaining and maintaining a travel speed, on level paved roads, of not less than 20 miles per hour (mph). The crane, without load, shall be capable of traveling on unimproved roads, rough terrain, and beach areas; shall travel up not less than a 30 percent grade on paved roads or hard-packed surfaces without engine stall; and shall be capable of negotiating 15 percent side grades.

3.9.1 Lifting capacities. The crane shall have lifting capacities of not less than the load specified, at the radius specified, and with or without outriggers set as specified in table I.

TABLE I. Minimum lifting load 360°.

Radius, FT	Load, LB	
	With Outriggers 360°	Without Outriggers 360°
10	30,000	19,500
12	26,500	13,000
15	23,000	9,500
20	15,000	6,000
25	10,000	3,200
30	7,000	2,500
35	5,500	1,750
40	4,500	1,200

3.9.2 Stability. With lubricating and cooling systems full and fuel tank(s) at least half full; crane resting on firm, level, and uniform supporting surface; outriggers unsupported, with the boom fully retracted and in travel position, the weight on the front axle shall be within 45 to 60 percent of the total weight of the crane. The crane, with boom fully retracted and placed at maximum boom angle without hook load, shall conform to the following requirements:

a. With the axis of the revolving superstructure at 90° to the axis of the carrier, the boom retracted and at maximum elevation, the total load on all wheels on the side under the boom shall be at least 15 percent of the working weight of the crane and carrier.
b. With the axis of the revolving superstructure on the axis of the carrier either forward or rearward, with the boom fully retracted and at maximum elevation, the total load on the wheels under the boom shall be at least 15 percent of the working weight of the crane and carrier.

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3.9.3 Power load lowering. Power load lowering by hydraulic control shall be provided for the main winch to facilitate controlled load lowering or lifting at any boom lengths or elevation. The power down shall have sufficient capacity to raise, hold, and lower a load of not less than 7,000 lb without slipping on a single part line. The power load lowering means shall provide positive control.

3.9.4 Rotation. The power transmission system used to rotate the superstructure shall be a hydraulic mechanical arrangement with a continuous rotation of 360°. Swing speed within this range shall be smooth and constant, and adequate control shall be provided to insure smooth starting and stopping. Swing speed shall be not less than 2 rpm. A swing lock or brake, capable of locking the superstructure in any desired position, shall be provided. The swing lock or brake controls shall be located within easy reach of the operator while seated.

3.9.5 Main winch capacity. The main winch shall have sufficient capacity to make maximum rated lift with the shortest boom length and with the length of rope installed on the winch specified in 3.10.4.

3.9.6 Safe load capacities chart. A permanent chart indicating all the safe load capacities on each boom reach and radii at 360° rotation, with and without outriggers, shall be posted in the crane cab near the operator and also on the outside of the crane to be legible from ground level. The safe load capacity chart shall reflect the manufacturer's standard commercial rating for the size and model crane furnished, and not be limited to the minimum required capacities specified.

3.10 Crane. The crane assembly, designed for the carrier mounting, shall consist of a fully revolving superstructure with boom support, hydraulically operated telescoping type boom, main winch with wire rope, hook block, and necessary accessories.

3.10.1 Superstructure. The superstructure shall be mounted on the carrier to insure 360° working range and designed to withstand the forces caused by lifting, swinging, and traveling with the safe loads. The superstructure shall provide stable retention of the boom in all positions and, with its base, shall rotate on ball thrust or roller bearings of sufficient number, size, and capacity to support working loads.

3.10.2 Boom. The boom shall be alloy steel and of the hydraulically operated telescoping type, consisting of a base section and sliding section(s). The boom tail swing shall not project more than 11 inches beyond the front and rear end of the carrier at any position of swing. For model I crane, the boom shall be able to be elevated and retracted from maximum radius position to a position with the hook directly over the centerline of the torpedo rack by hydraulic controls from the operator's cab. A boom angle indicator of the heavy pendulum type, swinging on sealed ball bearings, shall be provided. The pointer shall indicate the angle of the boom with the horizontal plane and shall be clearly visible to the operator. An anti two-block device shall be installed to prevent the hook block from contacting the boom tip.

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3.10.3 Hook block. The running block and hook shall have adequate weight, or a positive mechanical hydraulically actuated linkage, to insure payout of the hoist line without load on the hook when the hook is in any position. The hook shall be a swivel type with a safety factor of not less than four and shall include a safety latch. The hook shall not be painted. Block assembly shall be provided for attachment of hoist line for not less than two-part purchase. Means shall be provided to secure hook block while traveling. The hook block furnished shall permit disassembly for visual and nondestructive tests.

3.10.4 Rope. Load hoist rope for the main winch shall be not less than 6 x 19 improved plow steel, uncoated, preformed, independent wire rope center. The rope provided shall have a safety factor of not less than 3.5. The rope shall be not less than 450 feet in length and not less than 1/2 inch in diameter.

3.10.5 Sheaves. Sheaves shall have a pitch diameter at least 18 times the rope diameter. The hook block pitch diameter shall be at least 16 times the rope diameter. Sheaves shall be adequately guarded to prevent the wire rope from leaving the sheave during operation. All sheave grooves shall be smooth and the nominal groove diameter shall not exceed the nominal diameter of the wire rope furnished by not more than 0.1 inch.

3.10.6 Hydraulic system. The hydraulic system shall include a pump(s), hydraulic rams, rods, full-flow filter, reservoir, pressure-relief valve, and all necessary lines, lockouts, restrictor, and control valves to insure positive control of boom hoist and rope hoist in all operations or loss of hydraulic power due to ruptured hoses. Hydraulic line circuit and filtration shall be in accordance with SAE J931 with a 10-micron filter. The pump(s) shall be driven by the crane engine and shall have sufficient capacity to operate all hydraulically powered components as specified herein. Means shall be provided in the hydraulic system for bleeding all air trapped in the system. All rods, which will be exposed during operation, shall have a hard chromium plating with a crack-free thickness sufficient to withstand a 30-hour salt water spray test. All high pressure hydraulic hoses and fittings shall be capable of withstanding a bursting test pressure of four times the working pressure, and a maximum operating pressure of at least two times the working pressure.

3.10.7 Load-indicator and warning device. When specified (see 6.2), a load-indicating and warning device of the load-moment type shall be provided to perform the following functions while on outriggers:

a. Provide a dial or digital reading of hook load in pounds.
b. Provide a dial or digital reading of radius in feet.
c. Provide a dial reading of load-moment as a percentage of rated capacity.
d. Provide visual warning (amber light) when load-moment reaches 90 percent of rated capacity and visual and audible (red light and alarm) warning at 100 percent of rated capacity.

The device shall be capable of manually selecting parts of line and boom length. The device shall conform to SAE J159, J375, and J376.

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3.11 Cab. The cab shall be mounted on the carrier frame and shall be fabricated from sheet steel or fiberglass, having sufficient windows and glazed doors to permit at least 270° vision for the driver or operator in normal position. All glass shall be at least 7/32-inch nominal thickness, safety type, and held in place by rubber or metal channels for easy replacement. Windows shall be arranged for easy opening and removal when required. All cab doors, whether of the sliding or swinging type, shall be adequately restrained from accidently opening or closing when the crane is traveling or operating. All doors shall be provided with means for securely locking in the closed position. The model I crane cab shall be provided with a door, window, or windshield that is readily removable by the operator from inside the cab to provide an emergency exit when doors are blocked by torpedoes in the cradles and the boom is in travel position. An adjustable and comfortable seat shall be provided in the cab and located for maximum unobstructed visibility of the operator. The cab shall be equipped with heater, defroster, and air or electric operated windshield wiper. An electric or air horn, hand- or foot-operated, shall be provided. The cab shall be provided with outside rearview mirrors on each side. For the model 2 general purpose crane, a positive swing lock for traveling shall be provided in the cab.

3.12 Controls. All carrier and crane controls, lockouts, and indicators shall be located within easy reach of the operator in the operator's cab and shall be readily accessible under all conditions of operation. All controls shall be clearly marked as to their use and function.

3.13 Carrier. The carrier frame shall be of the heavy steel fabrication type, with necessary braces and reinforcements to provide rigid support for the revolving superstructure, hoisting drum rope, hook block, and cab, under maximum load, when the telescopic boom is extended at minimum through maximum radii at any position within the 360° rotation. The revolving superstructure shall be positioned in a location on the carrier frame that permits maximum efficiency for all-purpose crane operations. The carrier frame shall be mounted on four wheels with all wheels steerable. The crane shall have four-wheel drive. All wheels shall be equipped with interchangeable pneumatic tires of compatible sizes for the load imposed. Power-assisted steering for the front wheels shall be provided. When specified (see 6.2), power-assisted steering for the rear wheels shall be provided. When specified (see 6.2), indicators showing when rear wheels are centered shall be provided. Carrier shall conform to all applicable DoT Standards and Regulations.

3.13.1 Transmission. The crane carrier shall be equipped with a torque converter, a full power shift transmission with selective gear to provide not less than three forward and two reverse speeds. Gearing shall allow a maximum speed in the lowest forward and reverse gear of not more than 4.0 mph, and in the highest gear forward of at least the travel speed specified in 3.9 with the engine operating at engine-governed speed.

3.13.2 Brakes. The crane carrier shall be provided with the manufacturer's standard, foot-controlled, service brakes applied to all four wheels. The service brakes shall be capable of bringing the crane to a complete safe stop within a 50-foot distance on a level paved surface free from loose material from a speed of 20 mph. An independent parking brake, capable of holding the crane on a longitudinal grade of not less than 30 percent without slippage, shall be provided.

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3.13.3 Tires. Tires shall be of the tubeless type with traction type tread. Tires shall have a rated capacity at least equal to the load imposed on each tire, measured on each wheel at the ground, with the crane loaded with the maximum safe load. Tire ratings shall conform to T&RA recommendations for the type and size of tires furnished. When specified (see 6.2), tires shall be high flotation type.

3.13.4 Lights and reflectors. The following lights and reflectors shall be furnished and installed and shall be powered by a 12-volt electrical system:

a. Two sealed beam headlights (mounted on carrier).
b. Taillights.
c. Stoplights.
d. Turn signal lights.
e. Instrument panel light.
f. Emergency four-way flashing lights.

3.13.5 Engine. The carrier engine furnished shall be of the diesel type having horsepower, torque, and speed characteristics to meet satisfactorily all the crane performance requirements specified herein. The diesel engine shall start within 5 minutes and be ready for full-load operation within 15 minutes in any ambient temperature from -20° Fahrenheit (F) to +125°F. When an ether-priming system is required, it shall be of the measured shot type with storage capacity of at least 12 fluid ounces. The engine shall be furnished complete with at least the following accessories:

a. An instrument panel complete with a lubricating oil pressure gage, a fuel oil gage, a cooling liquid-temperature indicator, a battery charge and discharge rate indicator, a tachometer, and an hour meter.
b. A fuel tank with sufficient capacity for 8 hours of normal operation at 50 percent duty cycle.
c. A 12- or 24-volt electric cranking system.
d. Battery(s).
e. A charging alternator with a capacity rating compatible to the system.
f. A cooling-liquid high temperature safety device of the warning type to warn of high coolant temperature.
g. A lubricating oil safety device of the warning type to warn of low oil pressure.
h. A fuel oil level gage.
i. An engine housing so designed as to facilitate the maintenance of the engine.
j. A transmission temperature gage.

3.13.6 Axle assembly. The front and rear axles shall be of sufficient capacity to sustain the imposed loads with both axles. The rear axle assembly shall be of the oscillating type, with a positive-type oscillation lockout incorporated for crane operation. With lockouts engaged, the rear axle assembly shall not oscillate more than 2 inches measured at the centerline of the tires.

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3.13.7 Outriggers. Four hydraulically operated outriggers, capable of extension to 4 inches below ground level, shall be provided, one outrigger on each corner of the chassis. The outriggers shall be operated separately or in any combination from the operator's cab and shall not collapse due to loss of hydraulic power. The outrigger legs and pads shall be of a size to support the intended working loads without loss of crane stability or causing excess ground-bearing pressure. A leveling indicator shall be provided at the operator's station.

3.13.8 Front winch. When specified (see 6.2), a hydraulically operated winch of not less than 8,000-lb line pull rating on bare drum, shall be mounted on a frame extension at the front of the crane and protected fully against damage. The winch shall be of lightweight materials to the maximum practicable extent. The winches shall be equipped with an automatic brake. A device to prevent the winch from being inadvertently engaged shall be provided. The winch shall be wound with 150 feet of not less than 1/2-inch diameter wire rope, 6 by 19, Seale construction, improved plow steel with fiber center and with thimble and hook attached.

3.13.9 Pintle assembly. When specified (see 6.2), a pintle assembly shall be provided at the rear of the crane carrier. The pintle shall be installed on the carrier frame with reinforcement to transfer pintle loads directly to the carrier frame rails. The pintle type and pintle towing capacity shall be as specified by the procuring activity (see 6.2).

3.13.10 Toolbox. The crane shall be provided with a metal toolbox. The toolbox shall have a hinged lid with a padlock closing device less lock and key. The toolbox shall be securely fastened to the crane in a readily accessible and protected position.

3.14 Torpedo cradles. For model 1, the crane shall be equipped with two heavy gage metal trough-type cradles mounted lengthwise on the carrier. The cradles may be sectional to permit the cab doors to fully open when not carrying torpedoes. Both of the cradles are to have no end constructions. Each cradle shall be capable of carrying one 21-inch diameter torpedo, up to 20 feet in length and one or two 19-inch diameter torpedoes (depending on their length). The cradles shall be of a flat bottom, V-shape with 45° angle sides, approximately 9 inches high. Each cradle shall have hardwood or fiber centering and support blocks on which the torpedo(s) rest. Each support block shall be approximately 6 inches wide, 8 inches long, and 1 inch thick and each block shall have the torpedo-supporting surface covered with rubber (neoprene). Each support block shall be securely fastened to each side and bottom of the cradle at evenly spaced intervals (approximately 2-foot centers) throughout the entire length of the cradle. Each cradle shall be reinforced along its entire length and shall be capable of carrying a minimum of 5,000-lb distributed load.

3.15 Torpedo securing straps. For model 1, means shall be provided to secure/tie down the torpedoes in each cradle. Four nylon straps (1-3/4 inches wide) in accordance with MIL-W-4088, type 13, shall be provided with one end securely fastened to the side of the cradle and the other end to contain a tension buckle similar or equal to Part No. K761 or K711 with ACM/Cargo Ties, Inc., that will allow the strap to be adjusted and secured between 18- and

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30-inch diameters. The four straps shall be positioned in relation to the cradle length and insure a means of two straps when securing on 119-inch long torpedo.

3.16 Radio-noise suppressors. When specified (see 6.2), the crane shall be equipped with radio-noise suppressors conforming to SAE J552. Electromagnetic radiation from the crane shall be within the limits of SAE J551.

3.17 Lifting and tying down attachments. When specified (see 6.2), the crane shall be equipped with lifting and tying down attachments that shall conform to MIL-STD-209, type III. Attachments shall be indicated by a transportation plate. Tying down attachments may be identified by stenciling or other suitable marking on the crane. Tying down marking shall clearly indicate that the attachments are intended for tying down the crane on the carrier when shipped.

3.18 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173 except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.19 Identification plate. The contracting officer will furnish to the Government inspector the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit, assembly and/or subassembly, and parts as directed by the Government inspector. Nonferrous screws, rivets, or bolts of not less than 1/8 inch in diameter shall be used to affix the plates. For a Model 1 crane, the printed description shall be "CRANE, HYD., TORPEDO-HANDLING, 4 X 4, DED, CAP. 15 TON." For a Model 2 crane, the printed description shall be "CRANE, HYD., 4 X 4, DED, CAP. 15 TON."

3.20 Instruction plates. The crane shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material that will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.21 Stenciling. The gross weight of each crane with equipment shall be stenciled on each side of the crane in such a manner as to be readily discernible to dock and Military personnel. The prescribed tire pressure shall be stenciled on the carrier fenders by block or stencil-type letters not more than 1-inch high.

3.22 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. All parts requiring lubrication shall be properly lubricated as specified in section 5. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where

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use of high-pressure lubricating equipment, 1,000 pounds per square inch or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.23 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.24 Workmanship.

3.24.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions that would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.24.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.24.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.24.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subject to proof and service loadings.

3.24.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect that reduces the ability of the casting to perform its intended function.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be performed on one crane when a first article sample is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be a standard production item from the contractor's current inventory provided the crane meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining cranes to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.5, and the packaging inspection of 4.6.

4.3 Examination. Each crane shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests.

4.4.1 Crane safe load capacity test. The crane shall be completely serviced and subjected to the manufacturer's recommended break-in. The safe load capacity test shall be performed on a level, paved surface by connecting a Baldwin SR4 load cell, or equal, between the lifting hook and a dead weight or the direct use of calibrated weights. The crane shall then lift the safe loads specified in the least stable direction. The crane, with safe load

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raised, shall be rotated 360° clockwise and counterclockwise. Actual tipping loads, where a tipping load is the controlling factor, for each of the load conditions shall be determined in accordance with SAE J765. Eighty-five percent of each determined tipping load shall be compared with the applicable safe load.

4.4.2 Crane structure stress test. To determine conformance to 3.8, the working stress for crane structures shall be tested in accordance with SAE J1063. The certified test records shall be made available to the Government inspector and the contracting officer during first article testing.

4.4.3 Speed tests. Speed tests shall be taken over a 0 percent grade on a paved road. Time shall be recorded with a stopwatch accurate to 1 second. The average speed shall be recorded for the unit. Maximum average speed not less than that specified in 3.9 when calculated by the following formula shall be attained:

$$V = \frac{0.683 \times D}{T}$$

Where: V = speed in mph
D = distance in feet
T = time in seconds

A fifth wheel type recorder may be used.

4.4.4 Gradeability. The crane unit shall be tested for gradeability in compliance with 3.9. The crane, with mounted attachments and no load, shall negotiate a 30 percent grade forward and backward. The crane shall also traverse side grade of 15 percent.

4.4.5 Power load lowering test. General inspection of the power-down system shall be made for adequate lubrication, ease of operation of controls, safety precautions, and good manufacturing practices. The power load lowering system shall be tested as follows to determine conformance to 3.9.3:

a. With the crane equipped with a boom of a length as specified in 3.1, and with outriggers set, position the revolving superstructure in the most stable craning position and assemble a single line load equal to the maximum rated load divided by the recommended number of parts of hoist line reeving.
b. Position the boom angle for the minimum working radius.
c. Hoist the load 20 feet above the ground level, ±3 inches, or until interference with the crane boom is obvious.
d. Prepare the crane for power-down operation on the load line.
e. Operate the power load lowering system in accordance with manufacturer's instructions and lower the load under power. Vary the speed of the load.
f. When the load reaches the ground, immediately repeat operations c through e. These operations shall be repeated until 20 cycles have been completed.

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4.4.6 Crane hook test. The crane hook shall be given a wet or dry ferromagnetic particle test to determine surface or subsurface cracks or discontinuities. Either method of test used shall show clearly visible particles defining existing discontinuities. Any discontinuities shall be cause for rejection.

4.4.7 Load indicating device test. When the crane is equipped with the device, the load-indicating device shall be tested to determine conformance to 3.10.6.

4.4.8 Road test. The crane shall be given a road test for a distance of not less than 10 miles at a speed of 15 mph over paved roads, 4 miles over uneven roads with rough terrain at not less than 3 mph, and 1 mile over unpaved surface composed of sandy or loose soil or both.

4.4.9 Travel control test. To demonstrate workability of the travel controls (see 3.12), the crane shall be operated forward and backward in two figure-8 patterns with hard right and left turns.

4.4.10 Service brake test. Services brakes of the crane shall be tested for compliance with 3.13.2. On a concrete roadway, the crane shall be accelerated at maximum rate to a speed of not less than 20 mph. The service brakes shall be applied bringing the crane to a complete halt to determine conformance to 3.13.2. Immediately after the crane comes to a halt, this test cycle shall be repeated for four additional cycles. All shoes, drums, and other brake components where there is indication of damage shall be disassembled and examined.

4.4.11 Parking brake test. The parking brake shall be tested for compliance with 3.13.2.

4.4.12 Lifting and tying down attachment tests. When furnished, lifting and tying down attachments shall be tested to determine conformance to 3.17. The manufacturer may furnish certified engineering calculations in lieu of a test to prove that the lifting and tying down attachments offered meets the requirements set forth in MIL-STD-209.

4.4.13 Measurement of electromagnetic radiation. When required, to determine conformance to 3.16, electromagnetic radiation shall be measured in accordance with SAE J551. When suppressed to conform to 3.16, the manufacturer may, upon approval of the contracting officer, furnish a certification in lieu of the test that the crane meets the requirements, together with a list of the suppression devices installed. The list shall be sufficiently detailed to allow visual determination that the devices are installed.

4.4.14 Test records. Contractor's records of all first article tests and inspections, giving the results of said tests and inspections to determine compliance with the requirements of this specification, shall be kept complete and available to the Government's representative and the contracting officer. Test and inspection records shall be signed and approved by a person specifically assigned by the contractor. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a

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commercial testing laboratory, satisfactory to the Government, capable of conducting required tests to determine compliance with all the requirements of this specification.

4.4.15 Production sample. Upon acceptance of the first article, the first article shall remain at the manufacturing facility as a production sample and shall be the last crane delivered on the contract. The first article shall be reconditioned, including replacement of abnormally worn parts and paint touch-up or repainting, prior to delivery to enable it to be accepted as a contract item. The contractor shall maintain the first article in a serviceable condition for the duration of the contract.

4.5. Production unit operational test. Each production crane shall be completely assembled, adjusted, lubricated, or otherwise serviced for operation. The engine shall be started and subjected to a warmup period as recommended by the manufacturer. The crane shall be given a run-in test and all controls operated a sufficient number of times to ascertain that the components and mechanisms actuated by the controls operate promptly, fully, and without restriction or malfunction. Test for capacity loads specified in 4.4.1 shall be conducted.

4.6 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-C-3580. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-C-3580 with the level of preservation and level of packing as specified (see 6.2).

6. NOTES

6.1 Intended use. The intended use of model 1 is to load, transport, and unload torpedoes from cradles welded to the crane. Model 2 has a general-purpose design for loading and unloading torpedoes, ammunition, and electronic equipment to and from trailers and trucks.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Model of crane required (see 1.2).
c. When documents other than those specified herein are required (see 2.1.1).
d. When a first article is required for inspection and approval (see 3.2, 4.2.1, and 6.3).
e. When a load-indicating and warning device is required (see 3.10.7).
f. When power-assisted steering for the rear wheels is required (see 3.13).
g. When indicators showing position of the rear wheels is required (see 3.13).

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h. When high flotation type tires are required (see 3.13.3.).
i. When front-mounted carrier winch is required (see 3.13.8).
j. When pintle assembly is required (see 3.13.9).
k. When pintle type and pintle towing capacity is required (see 3.13.9).
l. When radio-noise suppressors are required (see 3.16).
m. When lifting and tying down attachments are required (see 3.17).
n. When fungus resistance is required (see 3.18).
o. When lubrication shall be other than that specified (see 3.22).
p. When cleaning, treatment, and painting shall be other than that specified (see 3.23).
q. Color of finish coat (see 3.23).
r. When a preproduction pack is required and the time frame required for submission (see 4.6).
s. Level of preservation and level of packing required (see 5.1).

6.2.1 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.3 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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